

Revised Load Interconnection System Impact Study Report

For Queue Project Q23-2277

Lake Hubbard Pump Station 138 kV
February 20, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2277. The Interconnection Customer proposes a 89.3 MW large load (data center) facility interconnecting at the Lake Hubbard Pump Station 138 kV substation in the New Jersey area. The request was received on January 15, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 688 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$881,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LAKE HUBBARD PUMP STATION / LAMB 138kV line	EAST KNOXVILLE / LAMB 138kV	115.4	215	43.7
PINETREE POWER TAMWORTH / EDISON 138kV	MILL C / LAMB 138kV	112.1	246	63.9
PINETREE POWER TAMWORTH / LAKE HUBBARD PUMP STA 138kV	MILL C / LAMB 138kV L	106.2	241	63.9
MILL C SUBSTATION / LAMB 138kV	N MESHOPPEN / KENNEDY 138kV	102.6	456	54.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LAKE HUBBARD PUMP STATION / LAMB 138kV line	Rebuild line with higher-capacity conductor	\$2,067,000	15% (\$309,000)
PINETREE POWER TAMWORTH / EDISON 138kV	Rebuild line with higher-capacity conductor	\$1,444,000	23% (\$335,000)
PINETREE POWER TAMWORTH / LAKE HUBBARD PUMP STA 138kV	Reconductor line	\$587,000	23% (\$136,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MILL C / LAMB 138kV	Reconductor line	\$739,000	14% (\$101,000)
	Total allocated to project		\$881,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.